

STEGANOGRAPHY



WHAT TO PROTECT

Level	What to protect	Method
3	Existence of message	Steganography
2	Metadata of message	Privacy-enhancing technologies
1	Content of message	Encryption
0	Nothing	None

Table by I. A. Goldberg

Metadata of message here is: the sender, the recipient, the time the message was sent, or the length of the message, etc

STEGANOGRAPHY AND INFORMATION HIDING



Steganography, derived from “covered writing” in Greek

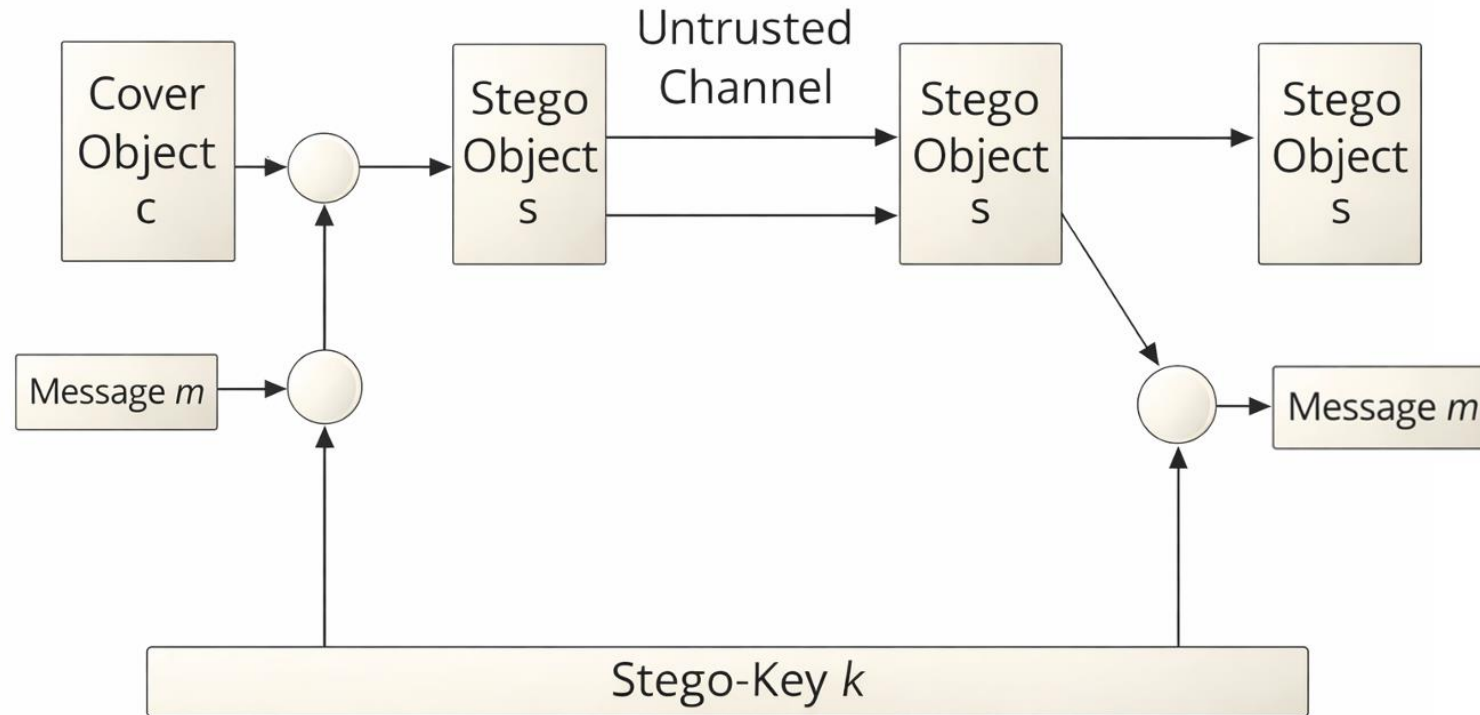


It includes the methods of secure communications that conceal the **very existence** of the message



Examples (non-digital): invisible ink, microdots, printer steg., etc

STEGANOGRAPHY IN DIGITAL WORLD



DIGITAL WATERMARKING

- Aim is to embed an amount of information that could not be removed or altered without making the cover object entirely unusable
- Adds additional requirement of **robustness** as compared **with** steganography
- Can be used for copyright protection

TEXTS AS COVER OBJECTS

Apparently neutral's protest is thoroughly discounted and ignored. Isman hard hit. Blockade issue affects pretext for embargo on by products, ejecting suets and vegetable oils.

(Real example of the text sent by a spy in WWII)

TEXTS AS COVER OBJECTS

*Apparently neutral's protest is thoroughly discounted
and ignored. Isman hard hit. Blockade issue affects
pretext for embargo on by products, ejecting suets
and vegetable oils*

Taking the second letter in each word gives the message:

Pershing sails from NY June 1

IMAGES AS COVER OBJECTS

LSB (Least Significant Bit) substitution method:

- Least significant bits used to store characteristics of particular pixels of an image (cover object) are modified to store a message
- Consider RGB 24-bit bitmap format
- Colours and brightness of pixels of obtained image may differ slightly from original cover image, but both images look identically to human eye
- Easy to implement, but not too robust methods
- Transformations of images may easily destroy the message (watermark)



IMAGES AS COVER OBJECTS



Watermarked image (LSB substitution)



Watermark recovered

Note: watermark is embedded as the image, not the plain text to improve robustness

ADVANTAGES AND DISADVANTAGES OF LSB

Advantages of LSB

easy to implement

has high capacity

Disadvantages of LSB

is not robust

message is easy to detect:

A message insertion introduces distortion to the *statistical properties* of image which never naturally appear

STOCHASTIC MODULATION METHOD

Simple variant:

- Before embedding a message, randomly chosen pixels are altered by changing their intensities (= a number between 0 and 255) by +1 or -1
- For a parameter $p \in [0,1]$ a pixel intensity is increased/decreased by 1 with probability p ; it is left unchanged with probability $1 - 2p$
- Then LSB method is used

Provide more protection against detection of the message.

STOCHASTIC MODULATION METHOD

Improved method (J. Fridrich, M. Goljan):

The idea:

- take a cover image and add a “noise” modulated by a message bits
- “noise” actually means pseudo-noise here, that is a sequence of pseudo-random values, which can be generated deterministically given a secret initial value (key)
- If initial value (key) is known, then generation of pseudo-noise can be repeated (used for extraction of the message)

STOCHASTIC MODULATION

- Simple implementation
- High capacity
- Low embedding and extraction complexity
- Embedding noise can have arbitrary characteristics and may approximate the noise of a given device => high security



TRANSFORM SPACE ALGORITHMS

- **Jsteg** algorithm (D. Upham) uses specifics of JPEG image format
- For each colour component JPEG image format uses *discrete cosine transform* (DCT)
- DCT is used by JPEG to transform 8 by 8-pixel blocks of the image to 64 DCT coefficients each:

$$F(u, v) = \frac{1}{4} C(u) C(v) \sum_{x=0}^7 \sum_{y=0}^7 f(x, y) * \cos\left(\frac{(2x+1)u\pi}{16}\right) \cos\left(\frac{(2y+1)v\pi}{16}\right)$$

where $C(x) = \frac{1}{\sqrt{2}}$ if $x = 0$ and $C(x) = 1$, otherwise.

TRANSFORM SPACE ALGORITHMS

- JPEG: after quantization DCT coefficients are stored
- Jsteg algorithm:
 - Replace sequentially the least-significant bit of discrete cosine transform (mid frequency) coefficients with the message data
- Gives better protection (as many others TS algorithms) against visual attacks

AUDIO (VIDEO) FILES AS COVER OBJECTS

- Temporal object. LSB can be used, but it introduces a significant noise to audio data
- A message may be encoded in audio signal *phase*, replacing original phase with a reference phase representing a hidden message; more difficult to implement
- *Spread spectrum* method: encoded data spread across the maximum range of frequencies; difficult to detect hidden message
- Video objects (files, streams) can be used for hiding information as well

NETWORK PACKETS AS COVER OBJECTS

- Steganography within TCP/IP:
 - Insert data within TCP and IP protocol headers
 - IP identifier, TCP initial sequence number, least significant bit of TCP timestamp, IP flags
 - Relatively easy to detect naive embedding by anomaly detection in TCP/IP fields
- One can prevent easy detection by taking into account the properties of concrete implementations of TCP/IP (Murdoch, Lewis, 2005)
- HTML steganography (PhD project 2015 – 2018)

REDUNDANCY

Steganography is applicable to any data objects that contain redundancy

Redundancy is used to hide the presence of the embedded message

On the other hand, redundancy may be removed during data compression

One may combine data compression and message embedding:
MP3stego by F. Petitcolas